

Independent Study

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This Week

- Highlights from my literature review marathon

AI Environmental Effects and Costs

Paper [One](#): How Hungry is AI?

This paper showed me that we must heavily consider how AI affects our environment. Furthermore, it provided the necessary measurements to calculate environmental effects, which I can replicate as closely as possible:

- PUE: non-computational overhead energy like lighting, cooling, etc.
- WUE: Water used per kw hour of IT energy
- CIF: Carbon emissions per kwh of energy consumed

A primary limitation this paper introduces with developing cost and environmentally efficient AI is Jevons paradox: If AI is cheaper and more efficient, then AI use goes up. Almost, leaving the environmental benefits to a net negative.

[Website](#): Cost calculations

We will use this OpenAI website to calculate costs per input/output token accurately. Furthermore, OpenAI's API provides more detailed information on how many tokens are spent on each task and how much they cost (e.g., input, reasoning, and output). This can also be helpful for calculating environmental overhead from the paper above.

AI Reasoning

Paper [two](#): Towards Large Reasoning Models

This paper explains how LRMs started and where they are today. This paper explains how COT is the first version of "thinking" LLMs. It also described training methods that use RL.

What stood out to me was how recent research shows AI agents being used in reinforcement search on inference time. In this search there are methods where they are able to have physical interactions and use tools. I wondered what would happen if we used these agents on neurosymbolic pipelines that could access semantic parsers as tools. This could genuinely make every neurosymbolic workflow smooth.

Conclusion

This was extremely useful. It seems as though methods improve very quickly in this day and age. While this gave me great ideas that I might focus on later, I should keep working on my project and hopefully achieve the results I want ASAP.

References

Papers and websites:

Any referenced papers and sites are linked inline above